Evaluation of the Event Horizon Fund (AI Infrastructure Portfolio) – January 2026

Overview of Portfolio Thesis and Allocation

The **Event Horizon Fund** is a specialized portfolio (~50% of total investments) focused on the AI infrastructure build-out. It concentrates capital in the **critical physical bottlenecks** of AI – areas where surging demand is constrained by hard supply limits or slow scale-up (e.g. advanced chip fabrication, wafer fab equipment, high-bandwidth memory, data center power & cooling)[1]. By doing so, the fund intentionally **avoids over-concentration in hyped end products (like GPUs)** and instead targets the upstream choke-points with durable pricing power as AI demand "collides with physics and policy"[1][2]. In practice, this means heavy emphasis on chip manufacturing and tooling, power infrastructure, and other enabling tech, rather than simply owning only AI chip designers.

Current Allocation (excluding the un-funded Moonshot sleeve) is spread across several key sectors and companies (with target percentage of the **AI sleeve** in parentheses):

- **Foundry & Advanced Packaging (12%)** – *TSMC (TSM)* – Leading-edge chip fabrication and advanced packaging (CoWoS).
- **Accelerators & Custom Silicon (9%)** – *NVIDIA (NVDA)* at 6%, *Broadcom (AVGO)* at 3% – AI GPUs and custom AI ASICs.
- **HBM Memory (6%)** – *Micron (MU)* – High-bandwidth memory for AI workloads.
- **Wafer Fab Equipment – WFE (12%)** – *ASML* at 8%, *Applied Materials (AMAT)* at 4% – Semiconductor capital equipment.
- **Electronic Design Automation (5%)** – *Synopsys (SNPS)* – Chip design software (EDA) tools.
- **AI Data Center Networking (5%)** – *Arista Networks (ANET)* – High-speed switching for AI clusters.
- **Rack-Scale Integration & Cooling (13%)** – *Vertiv (VRT)* at 8%, *Dell Technologies (DELL)* at 5% – Data center power/cooling systems and integrated AI servers.
- **Energy / Power Generation (18%)** – *Constellation Energy (CEG)* at 15%, *Quanta Services (PWR)* at 3% – 24/7 carbon-free power (nuclear) and grid infrastructure.
- **AI-Native Security (2%)** – *CrowdStrike (CRWD)* – Cybersecurity for AI-era threats.
- **AI-Native Operations (2%)** – *Palantir (PLTR)* – AI-enabled decision & control software.
- **AI-Enabled Defense (2%)** – *AeroVironment (AVAV)* at 1%, *Kratos Defense (KTOS)* at 1% – Autonomous systems and defense drones.
- **Bio-Risk Hedge (1%)** – *Moderna (MRNA)* – mRNA platform as insurance against AI-enabled bio-threats.

- **Strategic Materials (3%)** – *MP Materials (MP)* – Rare earth minerals for high-tech supply security.

Rationale: This allocation reflects the thesis that **the "hard infrastructure" of AI – chip fabs, fab tools, advanced memory, power grids, cooling, etc. – will accrue outsized value** as AI deployment accelerates. These segments have high barriers to entry or oligopolistic dynamics, allowing pricing power when demand vastly exceeds supply[3][4]. By contrast, the portfolio keeps only a **modest direct exposure to AI chip designers (like NVidia)** to avoid being the "marginal buyer of scarce GPUs" in a frenzy[5]. Instead, it backs the less appreciated upstream players that *all* AI efforts depend on (for example, owning the **sole-source suppliers** of EUV lithography, CoWoS packaging, HBM memory, or nuclear power needed for data centers). This approach provides leverage to the AI boom while mitigating the risk of overpaying for the most crowded trades.

Below, we evaluate each segment of the portfolio against current (2025–2026) industry conditions and trends, and recommend whether to **maintain, increase, reduce, or adjust** positions for maximizing future returns given the 2026–2028 AI outlook.

AI Infrastructure Trends (2025–26): Demand Outpacing Supply

AI compute demand is in an explosive growth phase, putting unprecedented strain on tech infrastructure worldwide. The market for AI chips alone has been growing ~30%+ annually – valued around **$53 B in 2024 and forecasted to reach ~$296 B by 2030**[6]. Training state-of-the-art models requires tens of thousands of GPUs, and enterprises and cloud providers are racing to build **massive AI datacenters**. Hyperscalers (like Google, Microsoft, Amazon, Meta, OpenAI) are investing **hundreds of billions** in AI hardware clusters, with some analysts projecting nearly **$7 trillion in cumulative global AI infrastructure spending by 2030**[7]. This **surge in demand is colliding with physical bottlenecks** in several areas:

- **Chip Fabrication Capacity:** The most advanced semiconductor fabs (≤5 nm) are effectively fully booked by AI chip orders. AI is poised to **consume the *entire* output of leading-edge fabs by the end of the decade** – forecasts show the share of sub-5 nm wafers used for AI skyrocketing from ~2.5% in 2023 to ~99% by 2030[8]. Simply put, virtually all cutting-edge chip production will be needed for AI if current trends continue. Yet adding new fab capacity is slow (multi-year build times) and extremely costly. This structural supply-demand gap gives enormous pricing power to the key foundry (TSMC) that controls this capacity[9][10].

- **Advanced Chip Packaging & Interconnects:** It's not just lithography – packaging technologies like TSMC's CoWoS (Chip-on-Wafer-on-Substrate) are a **governor on AI accelerator supply**. CoWoS capacity is ramping fast (~71% CAGR) but still lags the **80%+ annual growth in AI chip demand**, ensuring a persistent shortage of advanced packaging through the decade[11][10]. This has already manifested as delays/backlogs for HBM-based AI GPUs. Companies controlling these packaging

pipelines (again TSMC) have a de facto monopoly chokehold on high-end AI chip throughput[12][13].

- **Memory (HBM) and Networking:** Feeding and connecting giant GPU clusters is another friction point. High-Bandwidth Memory (HBM), which sits on AI accelerators, is in tight supply – only three suppliers make it, and one (SK Hynix) currently holds ~**57% share** while one (Samsung) stumbled to ~27%, allowing Micron (~16%) to rise as a new contender[14]. Similarly, traditional network technologies are being pushed to their limits by AI. The industry is rapidly shifting from proprietary interconnects (InfiniBand) to ultra-fast Ethernet fabrics for large-scale AI clusters. **Ethernet is now displacing InfiniBand as the preferred cluster interconnect**, and Arista Networks – the leader in high-performance switching – is at the center of this transition[15]. Demand for 400G and 800G switches has spiked as cloud players retool their data centers for AI workloads.

- **Datacenter Power and Cooling:** Perhaps the most dramatic constraint emerging is **power**. Training AI models draws **enormous electricity** – a single advanced AI datacenter can require 100 MW or more. Projections suggest **50 GW of new AI-related power demand in the U.S. by 2027**, which is ~3–4% of the entire U.S. generating capacity[16]. Globally, data center power usage could rise 165% by 2030 (vs 2023), with AI workloads' share of that doubling (from ~14% of data center power in 2023 to ~27% by 2027)[17]. This is a *huge* shock for electric grids that barely grew over the past decade[18]. The **only viable near-term ways to supply 24/7 power at this scale** are tapping existing nuclear and gas plants (since building new reactors or significant renewables+storage takes many years)[19][20]. Tech companies have begun locking in long-term power deals – notably, Meta (Facebook) signed a **20-year contract for nuclear power** from Constellation to run its AI datacenter[21][22]. Likewise, cooling these GPU-packed racks has become critical – **liquid cooling** solutions are now often mandatory as air cooling reaches its limits. The market for data center liquid cooling is projected to grow ~5× by 2034 (19%+ CAGR)[23]. **Vertiv** and others are racing to equip new AI mega-clusters with liquid cooling distribution units, rear-door heat exchangers, and coolant pumps. In fact, Vertiv has aggressively scaled up production – a *45× increase* in its liquid cooling manufacturing capacity in 2024 – to meet the pipeline of orders for next-gen AI facilities[24][25].

Bottom line: The AI boom is straining every link of the tech supply chain. This validates the Fund's focus on companies that *enable* AI at scale: **those who build "the factories that build AI," or power it, or cool it**. These upstream segments are seeing **unprecedented demand with limited ability to rapidly increase supply**, creating conditions for potentially extraordinary pricing power and profit growth in coming years. With that context, we now examine how each portfolio segment is positioned and whether the current allocation remains optimal.

Foundry & Advanced Packaging – **TSMC (12%)**

Taiwan Semiconductor Manufacturing Co. (TSMC) remains the *irreplaceable foundation* of the AI hardware ecosystem. As the world's largest contract chipmaker and the only source of leading-edge (<5 nm) chip fabrication in volume, TSMC is effectively the **physical gatekeeper of AI progress**[26]. *Every* high-performance AI accelerator (GPUs, TPUs, AI ASICs) ultimately relies on TSMC's advanced nodes and 3D packaging (CoWoS) to be manufactured. The fund's thesis identified TSMC as the "singular chokepoint" for AI chips[27], and that has only been **reinforced in 2025**:

- **Technological Lead:** TSMC's process technology and yields are well ahead of rivals. Samsung's competing 3 nm node has reportedly been stuck around ~50% yield, versus *near 90% yields* for TSMC's 3 nm process[28][29]. This huge gap is causing even Samsung's past customers (Qualcomm, etc.) to send more orders to TSMC[28]. Intel's foundry effort (IFS) is even further behind – still 1–2 generations back and mainly serving Intel's own chips[30]. In short, **none of the alternative foundries can currently match TSMC's reliability and output at the leading edge**.

- **Market Share & Pricing Power:** As a result, TSMC's dominance has grown. In Q2 2025 TSMC captured *over 70%* of the entire global foundry market – a record-high share[31]. It earned $30B+ in that quarter alone, mainly on robust orders for smartphone SoCs and AI GPUs[32]. With its near-monopoly on cutting-edge capacity, TSMC has been able to command premium pricing and require pre-payments for capacity reservations. The **imbalance between AI chip demand (forecast ~80% CAGR) and <5 nm wafer supply (~16% CAGR)** over the coming years means TSMC can essentially sell every wafer it can produce at elevated margins[4][10]. This structural tightness extends to advanced packaging (CoWoS), where TSMC also has a monopoly – CoWoS capacity, even expanding rapidly, remains a fraction of exploding demand[11][13]. All this gives TSMC durable pricing power; as the **sole supplier** of key AI manufacturing capacity, TSMC can raise prices or allocate output to its most strategic customers.

- **Virtuous Cycle:** TSMC's huge cash flows are being plowed back into R&D and new capacity (e.g. its upcoming 2 nm and 1.4 nm fabs). This scale and reinvestment, combined with learning-curve effects from massive volumes, further widen its lead. Meanwhile competitors stuck with low yields have less revenue to invest, a **"widening moat" dynamic**[33][34]. In short, TSMC's position as *the* leading-edge foundry for AI is more secure than ever in the mid-term.

Outlook: Maintaining a high allocation to TSMC (12%) appears well justified. The company sits at the nexus of the AI supply bottleneck – essentially tolling the AI revolution. Its recent results and market share gains confirm it's capturing the lion's share of AI-related chip manufacturing[31]. As AI model deployments scale 10×, TSMC should continue to see heavy demand and pricing leverage. **Key upside drivers** include

hyperscalers' multi-year AI chip roadmaps (more wafers for GPUs/ASICs), successful ramp of new nodes (3 nm, 2 nm), and expansion of packaging capacity to relieve the current CoWoS bottleneck[13].

Risks: The primary risk remains **geopolitical** – TSMC's production is concentrated in Taiwan. Any disruption (natural disaster or geopolitical conflict) would be a severe blow to the AI supply chain. The portfolio must accept this risk to obtain TSMC's outsized reward (and partially hedges it via U.S./Japan fab tool makers). Another risk is **Intel's future foundry ambitions** – if by 2027–28 Intel or Samsung were to significantly narrow the gap, TSMC's dominance could erode. At present that risk seems low (they trail by years)[30][35], but it warrants monitoring.

Recommendation: Keep TSMC at ~12% as a cornerstone. It continues to exemplify the fund's thesis: a non-substitutable bottleneck with pricing power in the AI era. So long as AI chip demand outpaces fab supply – which all data indicates will persist for years[4][10] – TSMC's profits should grow robustly. We uphold a **high-conviction, long-term stance on TSM**. (Investors should remain aware of the Taiwan risk; consideration could be given to diversifying a small portion into onshore foundry capacity if it emerges, but currently no comparable alternative exists.)

Accelerators & Custom Silicon – **NVIDIA (6%) and Broadcom (3%)**

These positions give the portfolio exposure to the **AI accelerator "arms race"** – principally through **NVIDIA**, the dominant GPU supplier, and **Broadcom**, a diversified chip company with a growing custom AI silicon business. The fund intentionally kept this sleeve at <10% to avoid overpaying for the most obvious AI winners; however, it's still important to own some of the upside from AI chip volume growth.

- **NVIDIA (NVDA)**: NVIDIA remains **the undisputed leader in AI accelerators**, with an estimated **80–90% market share** in data center AI chips[36][37]. Crucially, its moat extends beyond hardware – NVIDIA's *CUDA software ecosystem* is deeply entrenched as the standard for AI development[38]. This software lock-in means many AI researchers and enterprises are essentially "married" to NVIDIA's platform; switching to alternatives (whether custom ASICs or competitors' GPUs) carries high costs in re-tooling and performance risk. NVIDIA's latest financials underscore its dominance: for example, in FY2025 it saw **record data center revenues** (Q2 FY2026 was up 56% YoY to $46.7 B, driven by AI demand)[36]. The company continues to launch new architectures at a rapid clip (H100 in 2023, upcoming *Blackwell* GPUs in 2025, etc.), keeping performance leadership. Given the insatiable demand, **NVIDIA has been effectively selling every accelerator it can make**, and customers still face months-long lead times for H100 GPUs.

  Competition: Notwithstanding NVIDIA's dominance, the competitive landscape is **gradually evolving**. **AMD** has emerged as a credible #2 supplier for AI GPUs – its MI300 series accelerators (especially the MI300X) have started gaining traction in 2025, and analysts project AMD could capture ~15–20% of data center AI

accelerator share in a few years[39]. Indeed, some hyperscalers (e.g. Meta, Microsoft) have begun buying AMD Instinct GPUs to diversify suppliers. Importantly, this *does not fundamentally undermine NVIDIA* – it actually **validates and expands the market**. Hyperscalers prefer to avoid single-supplier risk, so having AMD as an alternative encourages *more* total investment in AI (a larger pie), of which NVIDIA still likely takes the majority[40][41]. NVIDIA's broad ecosystem (CUDA, software libraries, developer base) and superior all-purpose performance mean it remains the go-to for most cutting-edge AI training runs[42][43]. Meanwhile, specialized competition from **custom silicon** is also rising: Google has its TPUs, Amazon its Trainium/Inferentia for internal use, and notably **OpenAI reportedly partnered with Broadcom** to develop a custom AI chip (targeting 2026 production)[44][45]. These ASICs aim to optimize specific workloads (often inference or smaller models) at lower cost. However, they are largely used *in-house* by their creators and target narrower functions than NVIDIA's general-purpose GPUs. For frontier model *training* and third-party markets, NVIDIA's one-stop hardware/software solution still has no equal. The emergence of ASICs is actually a symptom of how pervasive AI workloads have become – big players are willing to invest billions to *augment* their NVIDIA deployments[46][47]. Through at least 2027, NVIDIA is poised to **dominate the high-margin "frontier" AI compute segment (training next-gen models and diverse enterprise AI)**, while custom chips handle some more cost-sensitive, standardized tasks (e.g. serving recommendations)[47]. In an exponentially growing market, this is not zero-sum; NVIDIA's slice can remain enormous even if select hyperscalers use some of their own chips.

Outlook: We recommend maintaining NVIDIA at around 6%. The company's execution and positioning continue to be stellar. It provides the portfolio a direct participation in AI compute growth **without** being over-weighted to it. NVIDIA's stock saw tremendous appreciation in 2023, so valuation is no longer cheap; but given its earnings are catching up fast (thanks to surging data center sales), the long-term thesis still holds. Key things to monitor are NVIDIA's ability to keep its software advantage (e.g. CUDA dominance), and its upcoming product cycles (the Blackwell and then *Vera Rubin* architectures)[48] – we need NVDA to stay on the cutting edge to justify its premium. So far it has, and rivals' advances (AMD, etc.) appear to be spurring **more** AI adoption rather than displacing NVIDIA[40][41]. Barring a dramatic change (like an unforeseen collapse of CUDA's relevance or a major stumble in chip launches), **NVIDIA remains a core AI beneficiary to hold**. If anything, the portfolio could consider **modestly increasing NVDA's weight** on significant dips – not because the thesis changed, but to ensure we capture sufficient upside from its earnings growth. We must balance this with the "don't be the marginal GPU buyer" philosophy; given NVDA's sustained performance, a slight increase (to, say, 8%) could be warranted if conviction remains high. Otherwise, holding at 6% and letting profits run is a sound strategy.

- **Broadcom (AVGO)**: Broadcom provides a complementary angle in this sleeve – it is a **diversified semiconductor company** that has become a critical supplier of **custom silicon for hyperscalers**. Broadcom's strength lies in its ability to design and deliver specialty chips (ASICs) under contract for cloud giants. For example, Broadcom's custom chip division is reportedly the partner for Google's TPUs and now OpenAI's upcoming in-house AI accelerator[45]. These lucrative deals leverage Broadcom's engineering without requiring Broadcom to shoulder the full product risk (the customer often funds much of the development). Beyond custom AI chips, Broadcom also dominates certain networking silicon (switch/router chips like Tomahawk series, NICs, DPUs) that are used in AI data centers. And recently, Broadcom acquired **VMware**, adding a software arm (though not AI-specific, it provides steady cash flows and enterprise reach).

  Recent performance: Broadcom has been a steady compounder; in 2023 it also saw an AI-related boost (e.g. its networking chips were in high demand for AI clusters, and it supplies critical components like specialized ASICs and switch chips to cloud customers). The **thesis for AVGO as a "diversified play on AI custom silicon" remains intact**[49]. In mid-2025, news of the $10 B Broadcom–OpenAI chip partnership reinforced that Broadcom is entrenched in the AI supply chain[45]. Financially, Broadcom has continued to grow earnings and maintains a reasonable valuation (especially compared to pure-play AI names). It also pays a hefty dividend. The presence of VMware now makes Broadcom a bit of a hybrid hardware/software firm, but its core semiconductor segments (networking, storage, custom silicon) are all seeing upside from AI build-outs (e.g. higher demand for cloud networking gear, and more custom chip projects as every hyperscaler considers their own AI ASIC).

  Outlook: We view **Broadcom's 3% allocation as a HOLD (potential slight increase)**. It offers stability (diverse revenue streams) plus targeted AI upside. Going forward, Broadcom should benefit from **the trend of hyperscalers "rebalancing" away from sole dependence on NVIDIA** – which often means either *partnering with Broadcom for custom chips* or buying more off-the-shelf networking silicon (which Broadcom leads). For instance, if Meta, Microsoft, Amazon all design partial in-house accelerators, Broadcom is likely involved in many of those. Broadcom's risk is that any single project can be lumpy (or if a cloud customer's chip underperforms, orders may not recur). But Broadcom's broad portfolio mitigates that. Given its reasonable valuation and ~8% earnings yield, one could argue **increasing Broadcom to ~4–5%** to weight it closer to NVIDIA. However, since Broadcom's AI exposure is only part of its business, keeping it a bit smaller is fine. The current 6% NVDA / 3% AVGO mix has worked well, capturing NVIDIA's high growth and Broadcom's stable returns. We will maintain that split for now.

  New Opportunity – AMD? A common question is whether to add **AMD** stock to directly capitalize on its rising role as the #2 AI GPU provider. At this time, we **recommend *not* adding AMD as a separate position**, for a few reasons. First,

AMD's success is already indirectly captured in our portfolio via TSMC (which manufactures all AMD chips) and perhaps Micron (if AMD-based systems drive more HBM sales). Second, AMD's AI story, while promising, is still developing – it has won some deals (rumored large orders from Meta, etc.), but it remains a fraction of NVIDIA's scale and faces execution risk in catching up. The portfolio's strategy is to hold the **clear bottleneck winners**; AMD, as a competing chip designer, doesn't own a structural bottleneck (it's actually *relying* on TSMC's bottleneck capacity). We also want to avoid over-diversifying into too many similar names. For now, **NVIDIA covers our primary GPU exposure**; AMD's emergence actually *boosts the overall AI investment level*, which helps many of our other holdings too[40]. We will keep AMD on our watchlist as its market share grows – if it appears that AMD will capture a very large portion of AI acceleration *and* the stock remains undervalued relative to that potential, we could reconsider. But at this juncture, **the incremental benefit of adding AMD is small** given our existing exposure, and it would dilute focus from the true bottleneck companies.

High-Bandwidth Memory (HBM) – **Micron Technology (6%)**

Micron (MU) is the portfolio's play on the memory side of AI – specifically the scarce **High-Bandwidth Memory** chips that are critical for GPUs and other accelerators. HBM acts as "ultra-fast RAM" attached to AI processors, and without enough HBM, even the best GPU cannot reach peak throughput[50][51]. The HBM market is a tight oligopoly (just three suppliers: SK hynix, Samsung, and Micron). Through 2024–2025, it has been one of the **most acute bottlenecks** in AI hardware. The fund chose Micron for this sleeve, and recent developments show Micron's position **strengthening**:

- **Oligopoly Dynamics:** As of mid-2025, **SK hynix commands ~57% of the HBM market**, thanks to its early partnership with NVIDIA (it supplied the majority of HBM2E/HBM3 for NVIDIA's A100/H100 GPUs)[14]. **Samsung's share collapsed to ~27%** – Samsung fell behind on the latest-gen HBM (its HBM3E was delayed in qualifying with NVIDIA), leading to **inventory write-offs of unsold HBM** and lost business[52][53]. This stumble by Samsung created an opening for **Micron**, the smallest player historically. Micron strategically skipped straight to HBM3E (bypassing the earlier HBM3 node) and executed **exceptionally well** – in fact, in early 2024 Micron became the **first to mass-produce HBM3E** memory, beating out both Samsung and SK hynix[54]. Micron's HBM3E was chosen for NVIDIA's newest Hopper-based GPU variant (the H200) shipping in mid-2024[54]. This was a major validation of Micron's technology.

- **Market Share Gains & Sold-Out Demand:** With Samsung lagging, **Micron is rapidly gaining share**. Analysts project Micron's HBM share will rise from the mid-teens (%) toward ~20%+, more in line with its overall DRAM market share[55][56]. Micron reported that **all its HBM production for 2025 was already sold out** well in advance, and much of 2026 capacity is booked too[56]. The company is also moving in tandem with SK hynix to develop even higher-density HBM (Micron is

gearing up 16-high layer stacks alongside Hynix)[55]. Essentially, Micron has shifted from being a negligible HBM player to an **integral supplier for top-tier AI customers** in a very short time. This is transforming Micron's financial profile – HBM and related high-performance memory carry **much higher margins** than commodity PC memory. Micron's mix is now moving toward specialized, oligopoly products (HBM, advanced server DRAM), which is **structurally improving its profitability**[57]. The market may still be valuing Micron like a typical cyclical memory maker, but Micron is becoming a key enabler of AI infrastructure (where spending is less cyclical and more secular-growth oriented).

- **Recent Performance:** Micron's stock had been weighed down by the broader memory downturn in 2022–23, but it bottomed in late 2023 and started recovering in 2024 as the DRAM/NAND glut eased. With the AI-driven **surge in HBM orders**, Micron's earnings are expected to rebound strongly in 2025. Each quarter of 2024 showed sequential improvement. Notably, Micron is one of few U.S.-based memory companies (which has earned it some subsidy support and strategic interest from the U.S. government, given China tensions). Micron's execution on HBM3E (hitting production milestones early) surprised many and has narrowed the tech gap with Hynix.

Outlook: We remain **bullish on Micron's role and thus keep the 6% allocation**. The "HBM gold rush" dynamic means Micron could enjoy several years of *sellers' market* conditions – demand far exceeds supply, giving pricing leverage. HBM content per GPU is also rising (new GPUs use more stacks), so even if GPU unit growth moderates, memory content growth supports Micron. Over 2025–27, Micron is positioned to ride the AI wave while also benefiting from a general memory cycle upturn (PC and smartphone memory markets are expected to recover, though more slowly). By 2026, Micron's earnings could be well above prior cycle peaks if HBM and advanced node DRAM stay hot.

Risks: Key risks include **technical execution** (memory is tough; any delays in Micron's next-gen HBM could slow its momentum), and the broader DRAM pricing cycle (Micron is still exposed to commodity DRAM/NAND swings, which could drag on results if, say, PC demand falters). Additionally, SK hynix will fiercely protect its lead – Hynix shipped the first samples of even newer HBM4 in 2025[58], so Micron must keep pace on R&D. Geopolitical issues are also a factor (e.g. China has at times banned some Micron products in retaliation to U.S. policies, though Micron's China exposure is limited). Lastly, **Samsung may recover** – Samsung has vowed not to miss the next cycle (HBM4), so by late-decade it could reassert share. But in the near horizon, Samsung is focused on catching up from a disadvantage.

Recommendation: Hold Micron at 6%. This position has significant upside if Micron's transformation into an AI memory leader continues. The allocation is appropriately moderate given Micron's historical volatility. If one wanted to be more aggressive (and didn't mind short-term swings), an argument could be made to **increase Micron's weight** a bit (to, say, 8%) to capture more of the AI memory upswing. However, it's prudent to keep

memory as a smaller slice than the more monopolisitic bottlenecks (like TSMC, ASML). We'll monitor Micron's HBM trajectory – so far, it validates the thesis that *memory is a crucial AI bottleneck and Micron is moving into a sweet spot*. No substitution is recommended; other HBM plays (SK hynix, Samsung) are either not easily investable or less attractive at the moment (Hynix is not U.S.-listed, Samsung is a conglomerate with many non-memory businesses). Micron remains the best pure-play for this theme in a U.S. portfolio.

Semiconductor Equipment (WFE) – **ASML (8%) and Applied Materials (4%)**

This sleeve targets the **"shovel-makers" of the semiconductor gold rush** – the companies that build the critical equipment for chip fabrication. If AI is driving a 10× expansion of chip demand, it implies a massive expansion of fabs worldwide, and thus a windfall for the **Wafer Fab Equipment (WFE)** makers. The fund holds ASML and AMAT here. Both have extremely strong positions and, importantly, the WFE industry structure gives them **durable pricing power and recurring revenues** – aligning well with the portfolio's bottleneck thesis.

- **ASML:** *ASML is arguably the single most critical company in the entire semiconductor supply chain*. It is the **sole provider of EUV lithography tools** – the machines needed to manufacture cutting-edge chips at 7 nm, 5 nm, 3 nm, etc. ASML's EUV scanners are technological marvels (costing $200–$300+ million each) with no competitors[59][60]. Every advanced logic chip – including all high-performance AI chips – **cannot be made without ASML's EUV machines**. This is the definition of a hard monopoly. ASML has ~90% share in overall lithography and 100% in EUV, and it has deep co-development partnerships with customers (TSMC, Intel, Samsung) that make it nearly impossible for a newcomer to catch up[59][60]. ASML's backlog of EUV orders has been huge, and it continues to ship more each year. Beyond selling new tools, ASML benefits from a **high-margin service business** – maintaining and upgrading these complex machines. Notably, ASML (and peers) now generate significant recurring revenue from service contracts, which can amount to *half of the tool's sale price over its life*[61]. This has smoothed out the cyclicality in ASML's business. With fabs running 24/7, service and spares are non-discretionary. In short, ASML is **less cyclical and more "infrastructure-like"** than past cycles assumed[61]. Financially, ASML has enjoyed strong growth and fat margins. As AI, 5G, and automotive drive more leading-node chip demand, ASML's EUV and upcoming High-NA EUV tools are in tight supply.

- **Applied Materials (AMAT):** Applied Materials is the largest overall WFE supplier by revenue, spanning multiple tool categories (deposition, etch, ion implant, CMP, etc.). It doesn't have a monopoly like ASML, but it's a leader in many segments and has a broad customer base across foundry, memory, and logic[62][63]. Notably, AMAT is a key provider of tools used in advanced packaging and in materials

engineering steps critical for AI chips (e.g. equipment for stacking memory, making through-silicon vias, etc.). It also has a substantial **services division (Applied Global Services)** that provides steady recurring income[64]. AMAT is well-positioned to benefit from the global fab expansion: new fabs in the U.S. (Intel, TSMC Arizona), TSMC's expansions in Taiwan/Japan, Samsung in Texas, Intel in Europe – **all will use Applied's tools**. Moreover, geopolitical developments (like U.S.–China tech tensions and the CHIPS Act subsidies) are causing **redundant fab build-outs in multiple countries**, which **multiplies equipment demand** (we're essentially building extra fab capacity for resilience, not just pure economic demand)[65][66]. AMAT, Lam Research (LRCX), and KLA are oligopolists in their respective niches. We chose AMAT for a broad exposure, but one could easily own any combination of these top U.S. WFE players.

It's worth noting: The portfolio does not currently hold **Lam Research (etch/deposition) or KLA (process control)**, but these are equally strong players in WFE. Lam, for example, dominates etching equipment needed for 3D NAND and for carving patterns in logic chips, and KLA's inspection tools are mission-critical for yield (and also have no close substitutes in function)[67][68]. All three – AMAT, LRCX, KLAC – plus ASML form a tight oligopoly. Given unlimited room, we might own all; but for manageability, the fund went with ASML (most unique monopoly) and AMAT (broadest product reach, large install base).

- **Recent trends:** 2023–2024 saw a downcycle in semiconductor capital spending for memory (due to the chip glut), but **leading-edge logic capex remained robust**. Now, with multiple new fabs being constructed (often government-supported), WFE orders are expected to pick up strongly through 2025–2026[65]. Export controls on selling advanced tools to China have created some headwinds (ASML can't ship EUV to China, for instance, limiting one source of demand). However, the **surge in Western and Asian (ex-China) fab investment more than compensates**, especially for advanced nodes. ASML's order book remains huge, and it's increasing tool output annually. Applied Materials, Lam, and others have noted that **geopolitical fragmentation is actually boosting their TAM** – instead of one mega-fab in Taiwan, we now see fabs in several regions, all requiring duplicate toolsets[65].

  A key shift in WFE is the focus on **long-term service revenue**. Investors historically saw WFE as cyclical (boom-bust with chip capex), but now the installed base is so large and the complexity so high that service is a growing, stable business. For example, ASML's service revenue and gross margins there are very high, helping cushion any new equipment slowdown[61]. Applied's services segment is its second-largest revenue contributor and helps stabilize earnings[64]. This means WFE companies might command higher valuations in the future than in past cycles, if the market appreciates this stability.

Outlook: The case for **WFE remains very strong**, aligning with the fund's view that they are fundamental bottlenecks (you can't build fabs faster easily; tool makers have long lead times). Our allocation is 12% (8% ASML, 4% AMAT). **We recommend maintaining this or even modestly increasing it** if conviction stays high. ASML in particular stands out as a must-own – it's an even "higher moat" business than TSMC in some respects, since **no one else can make EUV tools**[60]. We keep it at ~8% because of its relatively rich valuation and some cyclicality, but one could argue ASML deserves a double-digit percentage allocation given its monopoly. Applied at 4% is solid; we could consider **adding one of its peers (Lam or KLA)** by carving out a portion of AMAT's allocation, but that might complicate the portfolio without dramatically changing exposure. Alternatively, we might bump AMAT to 5% to reflect its upside as memory capex recovers (Lam Research has more direct memory exposure; since we don't hold Lam, AMAT gives some coverage there).

Overall, we are **bullish that WFE companies will enjoy a multi-year uptrend**. They sell the picks and shovels enabling the AI chip boom, and as noted in research, **each new fab built to hedge geopolitical risk basically duplicates demand for ASML/AMAT/LRCX/KLAC equipment**[65]. This is a secular tailwind.

Recommendation: Maintain ASML ~8%, AMAT ~4% (total ~12%). If there's any dip from short-term news (e.g., a quarter of slow orders), it would likely be a buying opportunity, as the medium-term drivers (AI, chip re-shoring) are intact. ASML should remain a top conviction hold – it's hard to imagine the AI 2027 scenario playing out without ASML's continued success[69]. We will watch for any changes such as: major delays in ASML's High-NA EUV (which might affect future tech roadmaps), or aggressive new entrants in any WFE niche (unlikely given high barriers). None of those seem imminent. Thus, **the current allocation to WFE is optimal**. We get both the high-growth monopoly (ASML) and the broad-based leader (AMAT) covering this critical bottleneck.

(Aside: The fund's internal analysis even suggested WFE might be a "more fundamental bottleneck than the foundries themselves," since tools are needed by every fab and have even fewer suppliers[70][59]. We concur that WFE is a vital area to be in. Any future increases *in exposure would likely go here (or possibly adding Lam/KLA), especially if geopolitical trends accelerate new fab construction.)*

Electronic Design Automation (EDA) – **Synopsys (5%)**

Synopsys (SNPS) represents the portfolio's stake in **chip design software (EDA)** – an often-overlooked but indispensable part of the semiconductor value chain. EDA tools are used to design and verify the complex ICs that companies like NVIDIA, AMD, Apple, etc. create. Synopsys (along with Cadence Design Systems) forms a duopoly controlling the majority of this market.

Thesis for EDA: No advanced chips can be built without EDA software – it's effectively the blueprint and simulation toolkit for chip engineers. As more companies jump into designing AI accelerators (including many new startups and cloud companies), demand for EDA tools and IP blocks rises. Importantly, EDA businesses have a highly attractive model:

software-like recurring revenue (licenses) with high margins, and they are not as capital-intensive or cyclical as chip manufacturing. The fund views EDA as an "upstream bottleneck" in that it is a **concentrated industry with huge barriers to entry** (the existing codebases and algorithms have been built over decades, and no newcomer has meaningfully challenged Synopsys/Cadence in core EDA in recent times)[71]. Also, Synopsys and Cadence have been embedding AI into their own tools (to handle the exploding complexity of chip design), making their offerings even more critical.

Recent performance: Synopsys has been delivering steady growth (~15-20% annually) and expanding margins. In 2024–25 it benefited from a wave of new chip startups (many funded to build AI chips) who all became customers for EDA. Big system companies (Amazon, Google, Tesla, etc.) designing custom silicon also license Synopsys IP cores and tools. Synopsys has a sizable IP portfolio (designs for standard interfaces, etc.) which gives it another revenue stream beyond software licenses[72][73]. The company's outlook has been consistently positive, with backlog and book-to-bill indicating strong demand. Cadence, its closest peer, has similarly strong trends.

Strategic position: EDA is something of a **quiet monopoly** – while not often in headlines, Synopsys and Cadence form a necessary duopoly that every chipmaker depends on[71]. This confers pricing power and durability. The **AI boom amplifies this** because AI chips are pushing the limits of silicon design – more chips, more complexity (billions of transistors, 3D stacking, etc.) which require even more sophisticated design and verification tools. Also, *new entrants designing chips* (outside traditional semiconductor firms) means the EDA customer base is expanding. Synopsys has noted "system companies" (like automotive OEMs, hyperscalers) are a growing segment of its clientele – a direct result of AI and edge computing trends.

Outlook: We believe **Synopsys (5%) is a solid hold, potentially even a candidate for a slight increase or the addition of Cadence** for a more balanced EDA exposure. However, 5% in one of the duopolists already captures the theme well. Both Synopsys and Cadence are highly correlated and have similarly high valuations (reflecting their quasi-monopoly nature). We chose Synopsys due to its slightly larger market share and broader IP portfolio[73], but frankly, either or both are good. The EDA segment gives the portfolio a relatively lower-volatility component that still rides the AI wave (since more chip R&D = more EDA spending). It also diversifies us a bit **away from heavy capital expenditure dependency** – EDA income is more subscription-like.

Recommendation: Maintain Synopsys at 5%. This provides diversified exposure to **the entire chip ecosystem's growth**, not just one company's chips[71]. In a way, owning SNPS (or CDNS) is akin to owning a *toll on all new chip designs*. The current allocation is fine. If the portfolio were enlarged, we could consider adding Cadence at maybe 3-5% as well, but that might be redundant for now. One thing to monitor: any regulatory changes or industry shifts (e.g. if open-source EDA tools ever became viable, or if customers push back on EDA pricing). So far, nothing significant in that regard – Synopsys and Cadence appear as entrenched as ever. Therefore, continuing to hold Synopsys is both a **defensive**

and offensive play aligned with AI's growth. It's robust, and we see no need to substitute or reduce this allocation.

AI Data Center Networking – **Arista Networks (5%)**

Arista (ANET) is our pick for the **network "fabric" of AI data centers**. In an AI cluster with tens of thousands of GPUs, high-speed networking is what links those GPUs into a single supercomputer. Arista specializes in exactly that: top-of-rack switches and routing systems with ultra-low latency and high bandwidth, plus an advanced network operating system (EOS). The portfolio's thesis anticipated that **Ethernet networking would become a bottleneck/opportunity as AI scales**, and indeed this has played out.

- **Ethernet vs. InfiniBand:** Historically, NVIDIA's Mellanox InfiniBand was commonly used for GPU interconnect due to its low latency. However, over 2024–25 there's been a strong trend toward using **standard Ethernet for large AI clusters**, because Ethernet has caught up in performance and offers more open, scalable architecture. Hyperscalers like Meta and Microsoft have signaled preferences for Ethernet-based AI networks, and Arista has been a big beneficiary[15][74]. According to industry analysis, **Ethernet is rapidly displacing InfiniBand as the preferred interconnect** for large-scale AI deployments[15]. Arista, being the technology and market share leader in high-performance cloud switching, has won a substantial portion of these new deployments. Its switches (running 400 Gbps and 800 Gbps speeds, with roadmap to 1.6 Tbps) form the backbone of many AI superclusters. For example, Meta's RSC AI supercomputer and Microsoft's Azure GPU clusters rely heavily on Arista gear (often using Arista switches coupled with Broadcom Tomahawk/Spectrum silicon inside).

- **Market leadership:** Arista has long been the top supplier to cloud giants for data center networking (outpacing Cisco in the high-speed segment). It holds a leading share in 100 GbE+ switching. In 2023–24 Arista saw **surging orders specifically for AI cluster builds** – this drove record revenue growth and backlog. The company noted that AI use cases were a major contributor to its >30% YoY growth in 2023. Arista's gross margins remain high (~60+%) indicating its product differentiation. The EOS software is a big part of its value – cloud customers can program and automate networks more easily with Arista's OS, which is increasingly optimized for handling **AI workloads and traffic patterns**[75][76].

- **Recent performance:** Arista's stock had a strong run in 2023 on the back of AI demand. By 2025, growth moderated slightly as some cloud customers digested their big 2023 purchases, but overall trend remains robust (cloud capex on AI networking is expected to rise again with the next wave of GPU cluster build-outs in 2024–26). Arista is financially very sound (no debt, strong cash flow) and is expanding into adjacent areas like campus networking (though its core data center business is the main AI play).

Outlook: Keeping Arista at 5% is recommended. It directly addresses the "data center fabric" bottleneck identified in the thesis. As AI models get larger and more distributed, the network becomes as crucial as the compute – without sufficient bandwidth, GPUs sit idle (network bottlenecks can reduce training efficiency significantly). Arista provides that **"central nervous system"** for the AI factory[77][78]. The secular shift to Ethernet is a tailwind in Arista's favor[76], and even NVIDIA has responded by introducing its own Ethernet-based offerings alongside InfiniBand (though NVIDIA's focus is more on bundling complete systems). Arista's independence and focus give it an edge with many cloud customers who want non-proprietary network solutions.

We see continued growth for Arista as 800G Ethernet adoption picks up through 2025–26 (driven by next-gen 5 nm switch chips, likely from Broadcom and Marvell, which Arista will package into systems). Potential **risks**: If NVIDIA's InfiniBand/NVLink approach makes a comeback or if proprietary interconnects (like optical mesh networks) emerge, it could slow Ethernet's march. So far, industry signals point to Ethernet winning due to flexibility and cost. Another risk is simply the lumpiness of cloud capex – Arista's big customers can dramatically alter order volumes quarter to quarter. But long-term, the trajectory is up.

Arista's position appears secure (its main traditional competitor, Cisco, has been slower in ultra-high-end innovation, and Arista has kept technological leadership in low-latency switching). New entrants are few – it's a tough market to break without incumbency.

Recommendation: Hold ANET at 5%. This allocation is appropriate for a high-growth, mid-cap company like Arista. If evidence mounts that *all* AI players are standardizing on Ethernet (versus, say, some clinging to InfiniBand), one could even argue for a higher allocation, because Arista would then address a larger share of AI infrastructure spend. However, 5% already captures the thesis well and balances risk (Arista, while a leader, doesn't have the absolute monopoly that some other picks do). No substitution needed – we picked the best-in-class operator in this niche. We'll monitor if Arista maintains its edge (e.g. design wins in next-generation AI cluster projects). Given their recent market share gains – within a rounding error of the #1 DC networking vendor[79] – we're confident in Arista's inclusion.

(Note: In earlier planning, Arista was on our watchlist pending confirmation of Ethernet's ascendancy. That confirmation has come – thus Arista was rightly moved into the portfolio[80][78]. It "perfectly complements" other holdings (NVDA for compute, Dell for systems, Vertiv/CEG for infrastructure) in building a complete AI data center[81].)

Rack-Scale Integration & Cooling – **Vertiv (8%) and Dell (5%)**

This category addresses the **practical deployment of AI hardware at scale**: assembling racks of servers filled with GPUs, and providing the power delivery and cooling solutions those racks require. The fund holds **Vertiv** (a leader in data center power/cooling equipment) and **Dell Technologies** (a leader in enterprise servers and integrated solutions). These two give complementary exposure to the **"physical build-out" of AI datacenters** beyond the chips themselves.

- **Vertiv (VRT)**: Vertiv has become a **picks-and-shovels play on AI datacenter build-outs**, particularly focusing on thermal management (cooling) and power distribution. As discussed, the **thermal load in AI computing is enormous**, and many facilities are shifting to liquid cooling. Vertiv is at the forefront of supplying liquid cooling infrastructure: coolant distribution units (CDUs), rack cooling doors, immersion cooling systems, as well as high-capacity UPS (uninterruptible power supplies), PDUs (power distribution units), and other electrical gear needed for dense computing racks. In the past year, Vertiv has seen **explosive order growth** from cloud and colocation providers deploying AI clusters. The Dell'Oro Group (analyst firm) noted Vertiv gained significant market share in data center power & cooling and is now virtually tied for #1 globally[82][24]. Vertiv's Q2 2025 sales were up 35%, and it amassed an ~$8.5 B backlog – much of that attributed to large AI-related projects[83]. The company responded by massively scaling production (e.g. **45× increase in cooling unit manufacturing capacity** in 2024) to meet demand[24][84]. Meanwhile, competitors are playing catch-up: Schneider Electric, one of Vertiv's main rivals, had to **acquire** a specialist (Motivair) because it was behind in liquid cooling tech[85]. This underscores Vertiv's **incumbency advantage** – it has design wins and deep integration with next-gen AI data center designs, which will be hard for lagging competitors to uproot[86].

  Financially, Vertiv turned around dramatically since 2022 (when it struggled with supply chain issues). Now it's posting solid profits, improving margins, and using its cash to reduce debt. Its stock reflected this, rising substantially through 2023. Even after that run, the valuation isn't stretched considering backlog and growth outlook. Essentially, **Vertiv is now the "entrenched incumbent" for cooling in many large AI facilities** – once their gear is spec'd in and installed, future expansions and maintenance also benefit Vertiv[83]. This gives a multi-year runway.

  Outlook for Vertiv: We remain very positive. The world's "AI factories" (like Microsoft's data centers, Meta's, Google's, etc.) are being built largely with Vertiv's infrastructure. The push for **higher efficiency and reliability** works in Vertiv's favor, since it offers advanced solutions in power and cooling. Also, if regulations or sustainability concerns force more liquid cooling adoption (air cooling is inefficient for high densities), Vertiv stands to gain even more. A risk is that **competition could eventually catch up** – e.g. Schneider integrating Motivair's products, or new entrants in immersion cooling – but given the scale and relationships Vertiv has, it's not immediate. We do note that Eaton (another competitor) is further behind in high-density cooling tech[87].

  Recommendation for VRT: Maintain or even consider increasing Vertiv (currently 8%). Vertiv aligns perfectly with the "bottleneck" theme: if liquid cooling becomes a mandatory element (which it effectively is for cutting-edge AI clusters), then Vertiv is akin to owning the picks & shovels of that mandate. We have it at a robust 8%, reflecting our conviction. Some might argue to boost it to ~10% given its momentum, but 8% is already a top tier allocation, and we should monitor

execution (ensuring they handle growth without stumbles). But certainly, **do not trim** – the thesis strength has only grown since initial inclusion.

- **Dell Technologies (DELL)**: Dell is a new addition (replacing an earlier position in Super Micro Computer) after analysis showed Dell's advantages in the **rack-scale AI server market**. Dell is the world's #1 server vendor overall, and it has moved aggressively into AI-optimized servers (offering systems pre-configured with NVIDIA GPUs, InfiniBand or Ethernet fabrics, and liquid cooling in some cases). In 2025, Dell's results demonstrated it has captured a **massive share of the AI server boom**. In the first half of FY2026, Dell shipped $10 B worth of AI servers (more than in all of FY25), and it raised full-year AI revenue guidance to $20 B[88][89]. In Q2 FY26, Dell's Servers & Networking revenue hit a record $12.9 B (up 69% YoY)[88]. Furthermore, Dell disclosed a staggering **order backlog of $11–14 B for AI servers**[89]. These numbers dwarf what smaller competitors (like Supermicro) can handle. **Dell's scale, supply chain, and enterprise relationships** make it the preferred partner for Fortune 500 companies and even many cloud builders when deploying AI at scale[90][91]. While early in the AI surge smaller players (e.g. Supermicro) grew fast by being nimble, the market is now about delivering tens of thousands of systems globally – a game that favors Dell's strengths in logistics, support, and financing[90][91].

  Dell also provides **end-to-end solutions** – it can bundle storage, networking, and services (important for enterprises that don't have large internal engineering teams like hyperscalers do). Dell's strategic partnerships (with NVIDIA, with Vertiv for integrated solutions, etc.) have further solidified its position. The portfolio's shift from Supermicro to Dell was driven by this recognition: the **"scale war"** of 2025–2027 will be won by those who can *deploy and support* AI infrastructure globally[90][91]. Dell's incumbency (58% overall server share vs Supermicro's ~6% in 2024[90]) gave it an insurmountable advantage once large customers moved beyond experimental phases to full deployments.

  Valuation-wise, Dell remains quite cheap (low double-digit P/E), partly because it's viewed as a mature PC/storage company. But the market may not yet be fully pricing the growth in its infrastructure solutions segment from AI. The stock also benefits from a substantial dividend and share repurchases.

  Outlook for Dell: We see **Dell as a stable, undervalued way to play the volume deployment of AI hardware**. It likely won't have the explosive upside of a pure-play AI stock, but it offers a great risk/reward profile. As AI adoption broadens to more enterprises (who will buy from OEMs like Dell rather than build in-house), Dell's pipeline should stay strong. The key risk is if commoditization pressures margins (if many competitors offer similar boxes, price competition could increase). However, Dell's scale and services have let it maintain decent margins. Also, if cloud providers decide to design more of their own servers (some do), Dell might miss

that, but cloud is already doing that and Dell still got a chunk of business (plus Dell targets the broader market beyond just hyperscalers).

Recommendation for DELL: Hold at 5%. The decision to include Dell has been validated by its recent performance. Five percent is an appropriate weight – it gives meaningful exposure but acknowledges Dell is a diversified company (only part of it is AI-driven). We would not increase it much beyond this, since it's more of a steady grower than a high-octane play. But we definitely wouldn't cut it either, as its role in the portfolio is distinct (covering the system integration aspect of AI hardware). Dell helps ensure the portfolio isn't missing the "last mile" of turning chips into deployed capacity. We will watch how Dell's AI order backlog translates into revenue over the next 1–2 years; if it consistently exceeds expectations, there could be a case to inch the allocation up. For now, maintain 5%.

In summary, **Vertiv and Dell together make a formidable duo covering AI data center infrastructure**. The current combined allocation (13%) seems well-calibrated. Vertiv captures the **infrastructure bottleneck** (power/cooling), and Dell captures the **deployment/scale bottleneck** (manufacturing and delivering complete AI systems). They both have shown they can execute at scale, and they round out the portfolio's hardware exposure beyond just chips.

Energy & Power Infrastructure – **Constellation Energy (15%) and Quanta Services (3%)**

The **Energy sleeve (18%)** is the largest in the portfolio, reflecting the thesis that **electric power will become the ultimate bottleneck in AI** once compute supply ramps. As discussed, an AI-driven surge in electricity demand is already underway, and **reliable, round-the-clock power** is a scarce resource. The fund holds **Constellation Energy (CEG)** as a pure-play nuclear power generator, and **Quanta Services (PWR)** as the leading contractor for grid expansion and transmission projects. This combination addresses both **power generation** and **power delivery**.

- **Constellation Energy (CEG)**: Constellation is the **largest nuclear power operator in the U.S.**, with ~21 GW of nuclear capacity (and some gas/other assets)[21]. Nuclear plants produce steady **baseload power** with >90% capacity factors – exactly the kind of 24/7 power that always-on AI supercomputers require[19]. The portfolio thesis posited that as AI datacenters demand huge and **uninterruptible** electricity, existing nuclear plants would be "re-rated" as strategic assets (because building new ones takes too long)[92][20]. This is playing out: tech companies have started securing nuclear-sourced energy via long-term PPAs. The standout example is **Meta's 20-year deal with Constellation** to supply a new datacenter with nuclear power[21][22]. This kind of deal effectively **monetizes CEG's nuclear capacity at premium rates** (likely above typical market power prices, given the reliability and carbon-free nature). It validates that **CEG's generation fleet is unique and highly valuable** in the AI era. Compared to alternatives, Constellation is a more focused

nuclear play than peers like Vistra – nuclear is ~70% of CEG's output vs only ~16% for Vistra[21]. That focus means CEG stands to benefit more from any nuclear renaissance or repricing[93]. Financially, Constellation has been strong since its spinoff (from Exelon) – it benefited from rising power prices in 2022, and more importantly from the **Inflation Reduction Act (IRA)** which introduced production tax credits for existing nuclear plants. Those credits (up to $15/MWh) effectively subsidize CEG's costs and ensure profitability even if market power prices dip. As a result, CEG has steady earnings and cash flow outlook through the 2020s, with upside if it signs more contracts like the Meta deal or if power prices rise due to demand.

Strategic outlook: Through 2027, **no significant new nuclear capacity will come online** (small modular reactors won't arrive until 2030s)[92]. So the existing reactors (like CEG's) are essentially *irreplaceable* for clean firm power. Meanwhile, data center power demand is skyrocketing[16][18]. This points to **continued tightening of power markets** in regions with lots of AI buildout (e.g. U.S. Mid-Atlantic, Midwest – areas where CEG has plants). If grids face shortages, the value of having an owned fleet of generating stations goes up. Also, many corporations have 24/7 carbon-free goals, which only nuclear (or hydro/geothermal) can fulfill around the clock – wind/solar alone can't due to intermittency[20]. This regulatory/corporate ESG push could drive more direct nuclear PPAs.

Recommendation for CEG: Maintain the overweight ~15% allocation. Constellation remains one of the highest-conviction positions. It provides a different kind of "bottleneck" exposure – not tech, but infrastructure – which adds defensive balance to the portfolio. Its stock performance has been solid, but not yet reflecting the full "AI power shortage" thesis (it's up from 2022 lows but still at a moderate valuation ~15× earnings). We expect as more deals and evidence accumulate (data centers literally unable to get enough grid power, etc.), investors will further re-rate CEG. The combination of stable earnings (thanks to IRA credits and capacity payments) with upside from AI-driven demand makes it attractive. Risks to watch: commodity price swings (if natural gas prices crash, power prices fall – though CEG hedges much of its generation), nuclear operational risks (unplanned outages can hit earnings), and policy changes (though current policy is supportive). None of these detract from the strategic need for CEG's assets. Thus, keeping CEG as the **single largest holding** is justified by our conviction that **"megawatts are the final gatekeeper" of AI deployment[94][16]**.

- **Quanta Services (PWR)**: Quanta is a **leading engineering & construction firm for electric power infrastructure** – essentially, the top contractor for building out transmission lines, substations, and other grid projects in North America. If AI is driving new power generation, that power also needs to be delivered to datacenters, many of which are in locations requiring **new transmission links or on-site high-voltage installations**. The thesis view is that **transmission is a co-equal bottleneck to generation** – it's not enough to produce electricity; you must also

have the wiring and substations to get it where it's needed[95][96]. Grid expansion in the U.S. has historically been slow and underinvested. Now, with both AI demand and renewables integration, there's a massive wave of required investment in high-voltage lines (the U.S. DOE estimates hundreds of billions need to be spent this decade). **Quanta Services is the go-to builder for this** – they have the workforce, expertise, and nationwide footprint.

In 2023–25, Quanta saw strong business from renewable energy tie-ins (lots of solar/wind projects to connect) and started to get more datacenter-related contracts (for instance, running new transmission feeds to hyperscale datacenters, upgrading local grids for increased loads). Quanta's backlog hit record levels, and it has been guiding high-single to double-digit revenue growth. Quanta also acquired companies in adjacent fields like telecommunications fiber and pipeline infrastructure, but power remains its core segment.

Strategic importance: If AI power consumption really becomes *double-digit% of grid load* by 2030 (which some forecasts suggest)[16], there will be enormous pressure to rapidly build out grid capacity – new lines from generation hubs to load centers, and upgrades to existing lines for higher throughput. Quanta will be tapped for many of these projects. It's essentially a **derivative play on grid modernization**, which itself is partly driven by AI (as well as electrification and renewables).

Recent performance: PWR stock performed very well in 2021–22 (driven by renewable grid buildout momentum), and after a pause, remains at high levels relative to pre-2020. Its valuation is not cheap (investors recognize its secular growth), but given the growth runway, it's reasonable. Quanta's earnings are less volatile than in the past due to more multi-year regulated utility projects (less seasonal swings).

Recommendation for PWR: Hold at least 3%, with a bias to add on any weakness. Currently 3% is relatively small, but we included Quanta more recently as we realized **not owning the "delivery" side of power would leave a gap in the thesis[95][97]**. Now that it's in, we think it's a keeper. One could argue for increasing it to ~5% to better represent the grid bottleneck, which truly is critical (in some areas, lack of transmission capacity is *already* delaying datacenter projects – e.g. Northern Virginia's data hub faced moratoria due to grid constraints). However, Quanta is a contractor, so its margins are slimmer and it faces execution risks on big projects (weather, labor, etc.). Thus we keep it a notch below our core tech holdings in size. But it's absolutely the right name to hold for this theme – Quanta is essentially without peer at scale (some smaller regional players exist, but Quanta is dominant). Given the U.S. government's push (CHIPS Act, infrastructure bill, etc.), regulatory support for grid build, Quanta's prospects look strong. We will monitor its project pipeline tied to datacenters specifically; if AI really stresses grids, utilities may accelerate transmission plans, benefiting Quanta.

In summary, **the Energy/Power sleeve remains crucial and is likely still underappreciated by many investors**. The portfolio's heavy 18% weighting here is a deliberate overweight because we foresee a scenario where compute is abundant but energy is the limiting factor. So far, data points (like Meta's nuclear deal, Texas grid strains from new data centers, etc.) are confirming this narrative. **We will maintain the current allocation, keeping CEG as a top holding and Quanta as a strategic smaller holding**, adjusting only if needed to reflect even greater conviction (e.g. if more tech companies follow Meta's lead, we'd double down on CEG).

(Footnote: Internal analysis concluded that evidence "overwhelmingly confirms power will be the final and hardest bottleneck," increasing conviction in CEG[98][99]. We agree – it's a high-conviction theme that differentiates this portfolio from typical tech portfolios.)

AI-Native Security – **CrowdStrike (2%)**

This small sleeve is meant to cover **cybersecurity in the AI era**, focusing on companies that *natively leverage AI and address AI-related threats*. **CrowdStrike (CRWD)** was chosen over more traditional security vendors because CrowdStrike's cloud-native platform and focus on endpoints/identity make it well suited for the emerging landscape of AI-driven attacks and autonomous agents. The portfolio's thesis highlighted that future cyber threats may involve AI systems themselves (rogue or malicious AIs exfiltrating data, or AI-assisted hacking at superhuman speed)[100][101]. In such scenarios, legacy perimeter defenses (like firewalls) are not enough – security must assume breaches and constantly monitor internal behavior[102][103]. CrowdStrike's approach, which uses lightweight agents on endpoints and cloud analytics powered by AI, is tailored for this "assume breach" model.

Why CrowdStrike: CrowdStrike has been a leader in using AI for threat detection for years. Its platform (Falcon) protects endpoints (servers, laptops, cloud workloads) by detecting anomalous behavior. In 2023–25, CrowdStrike started rolling out specific **AI security modules**, for example: tools to **discover and secure AI applications/LLMs running in environments**, scanners for vulnerabilities in ML packages, and an AI assistant ("Charlotte AI") to help automate cybersecurity tasks[101][104]. This shows CrowdStrike is directly aligning its roadmap with new AI-related risks. Its vision is essentially *"use AI to defend against AI"*, which our thesis agrees is the only scalable approach against future threats[105][106].

Financially, CrowdStrike continues to grow at ~20-25% annually (as of 2025) with improving profitability. It has been gaining larger enterprise contracts and expanding its product modules, leading to strong net retention. In contrast, a legacy player like **Palo Alto Networks (PANW)**, while still growing ~15%, is more focused on network firewalls and a broader platform. We specifically pivoted away from Palo Alto in favor of CrowdStrike because **the nature of AI-driven threats (insider threats, cloud-based attacks) renders perimeter firewalls less central**[107][101]. For example, if an AI agent inside a network goes rogue, a firewall doesn't help – but CrowdStrike's endpoint monitoring might detect unusual activity by that agent on a server[108][102]. CrowdStrike's pure-play focus means

it's innovating faster on these new use cases, whereas Palo Alto is balancing many legacy product lines.

Outlook: Security spending tends to be resilient and only growing as threats increase. As more AI systems are deployed in enterprises, those become new attack surfaces that need protecting. We anticipate **regulatory frameworks** will eventually require organizations to have certain security controls for AI (audit trails, data protection, etc.), which could create demand for products like CrowdStrike's AI-specific modules. Also, any headline cyber incidents involving AI will spotlight this theme.

At 2% allocation, CrowdStrike is a **small but strategic** part of the portfolio. Its stock is somewhat volatile (high valuation), so we kept weight modest. However, given the heightened geopolitical cyber tensions (with AI tools amplifying them)[109][110], one could justify increasing security exposure. In fact, our internal review suggested potentially **increasing CRWD's allocation on future rebalances** due to the critical nature of AI-era threats[111][112]. Even a move from 2% to ~3-4% would be reasonable as CrowdStrike continues to execute.

Recommendation: Consider a slight increase in CrowdStrike (from 2% to ~3%) over time, especially if we trim any non-core positions. The portfolio currently prioritizes physical infrastructure, but **cybersecurity is a form of infrastructure, too (digital infrastructure)**. CrowdStrike is arguably the "must-have" name in next-gen cyber defense for AI, akin to how we view CEG for power or VRT for cooling. That said, 2% still provides exposure, and if other weights are to remain fixed, it's acceptable. We definitely **advise against dropping CRWD** – that would leave a gap. If anything, better to boost it slightly. We will keep watch on competition (e.g. Microsoft's security offerings, SentinelOne, etc.), but CrowdStrike has maintained a technology lead and high growth relative to peers. With its strong alignment to our AI-security thesis, **CrowdStrike remains the optimal holding** here.

(In sum, the choice of CRWD over PANW was about precision: CrowdStrike's vision "perfectly matches the unique risks of the fast-takeoff AI thesis"[101][104]. We agree with that analysis and see no reason to revert – Palo Alto is great in traditional security, but for AI-specific threats CrowdStrike is ahead.)

AI-Native Operations & Control – **Palantir (2%)**

This sleeve is aimed at companies that help **operate, orchestrate, or apply AI in the real world** – essentially the software "glue" between AI models and large-scale real-world decision-making. **Palantir (PLTR)** is the standout candidate here. Palantir provides platforms (Foundry, Apollo, AIP) that allow organizations to integrate AI into their operations with proper data governance, monitoring, and control. The fund included a small position (2%) in Palantir as a play on **enterprise/government adoption of AI at scale**.

Palantir's role: Palantir is often described as offering an "operating system for the modern enterprise/government." Its software can ingest data from across an organization, structure it (the "ontology" layer)[113], and enable both human and AI agents to interact with that data in a governed way. This makes it extremely well suited to be the **control layer for AI implementations** – for example, a military using AI surveillance drones can use Palantir to integrate the AI outputs with other intel and decide actions, all while keeping humans in the loop. In an enterprise context, Palantir Foundry can help a company deploy AI models on their proprietary data and immediately operationalize the results (e.g. optimizing supply chains, detecting inefficiencies)[114].

Recent developments: In 2023–25, Palantir leaned heavily into AI – launching its **Artificial Intelligence Platform (AIP)** which allows organizations to harness LLMs on their private networks with control. They demonstrated use cases like having an LLM interact with classified military networks (but with guardrails) – something highly aligned with the AI control thesis. Palantir's deep ties with government (they have big contracts with DoD, for Army's TITAN program, etc.) give them a **trust advantage in high-stakes environments**[114][115]. At the same time, their commercial business saw growth as more companies reference Palantir in earnings calls as a key to their AI strategy[116][117]. In fact, mentions of Palantir on other companies' calls quadrupled, a sign of rising traction[116].

Financially, Palantir achieved its first profitable quarters in 2023 and has guided to continued profitability. The stock, which was beaten down in 2022, rallied in 2023 on AI hype (nearly tripling at one point). It's still seen as a bit speculative because its valuation anticipates high growth that needs to materialize. That's why we kept the allocation modest.

Outlook: Palantir is potentially a **big long-term winner if AI adoption in government and regulated industries takes off**, because few others provide the integrated, secure solutions that Palantir does. It could also be an acquisition target (though its governance likely prevents that) for a major tech firm wanting an inside track in gov AI projects. The key is execution – turning pilot projects into large recurring deployments.

Recommendation: Hold Palantir at 2%. We view it as a high-upside, moderate-risk bet on the "AI operating system" theme. Given its volatility, 2% is enough to contribute if it soars, but won't hurt the portfolio badly if it underperforms. We wouldn't increase it until we see more concrete evidence of accelerating revenue growth from AI products (the latest earnings did show some uptick, but not explosive yet). The fund's internal guidance was to **keep Palantir on the watchlist and potentially add on pullbacks**[118][119] – which we did earlier in 2023 around $8/share. Now it's in the portfolio; going forward we'll likely *add on dips* to maintain roughly this percentage. If the stock runs far ahead of fundamentals (as AI enthusiasm sometimes causes), we might trim a bit but currently it's at a reasonable size.

No substitution is recommended – there aren't many pure competitors. One could consider other AI software plays like C3.ai or enterprise SaaS firms, but none have

Palantir's combination of defense pedigree and enterprise traction. So Palantir remains the **best-in-class "AI ops" play** for our purposes. We'll look for continued signs that it's becoming integral to government AI efforts or that its AIP is gaining commercial customers. If positive, we hold; if it falters in delivering growth, we'd re-evaluate, but at 2% the risk is contained.

AI-Enabled Defense – **AeroVironment (1%) and Kratos Defense (1%)**

This sleeve provides exposure to the intersection of **AI and military/defense technology**, a theme that could see significant acceleration given global tensions and an AI arms race. We have two small positions here: **AeroVironment (AVAV)** and **Kratos Defense (KTOS)**, each ~1%. The idea is to capture the upside of defense systems that incorporate advanced AI/autonomy (like drones, robotic vehicles, smart missiles) without investing in large defense primes that have broader businesses.

- **AeroVironment (AVAV)**: AeroVironment is a leading manufacturer of small **unmanned aircraft systems (UAS)** – essentially drones used for reconnaissance and strike. Notably, AVAV's Switchblade loitering munitions (basically small kamikaze drones) have been used heavily in the Ukraine war. AeroVironment has a strong incumbency in **tactical drones for the U.S. and allies**, and it's positioning itself to be a dominant player in **AI-driven autonomous systems** for the military[120]. They've been integrating more advanced autopilots, computer vision, and AI target recognition into their drones. Their recent acquisitions and R&D focus on networking swarms of drones and deploying AI for target selection (with human oversight). The portfolio included AVAV because it's a pure play on the trend that warfare is shifting to smaller, smarter, autonomous platforms – a trend likely to accelerate as AI improves decision-making and autonomy.

  AVAV's financials got a boost from foreign demand (Ukraine orders, European rearmament). Longer term, programs like the U.S. Army's FTUAS (future tactical UAS) could drive growth, and AVAV is a top competitor there. The company's vision (as articulated by its CEO) is to become the **dominant provider of intelligent, networked robotic systems on the battlefield**[120][121]. If it succeeds, it could be much larger in a decade.

- **Kratos Defense (KTOS)**: Kratos is a smaller defense tech firm with special focus on **unmanned systems, target drones, and space/missile systems**. The key reason to hold Kratos is its work on **affordable autonomous jet drones** – notably the **XQ-58 Valkyrie**, a "loyal wingman" type unmanned jet that can fly alongside manned fighters, using AI to perform missions semi-independently. This is part of the Pentagon's vision for a future where many **cheap AI-powered drones augment expensive manned aircraft**. Kratos also makes high-performance target drones used to test U.S. fighters and air defenses, giving it relevant technology. The fund included KTOS as a speculative bet on the DoD's new "Collaborative Combat Aircraft" (CCA) concept – the USAF and USN plan to acquire *hundreds or thousands*

of autonomous drones to team with pilots[122][123]. Kratos, being a pioneer in this space with Valkyrie, could win some of these contracts. It's essentially an option on a **paradigm shift in military procurement** (quantity over pure quality, leveraging AI for swarms of lower-cost assets)[120][123].

Financially, Kratos is not yet consistently profitable and its revenue growth has been lumpy. That's why its weight is small – it's higher risk. But any major contract win in CCAs or other AI-enabled systems could dramatically increase its value (hence the asymmetric upside rationale in the Moonshot sleeve originally).

Outlook: The defense AI theme is *highly plausible* to become big by 2027, given geopolitical competition (US, China racing in autonomous weapons). The **recent spotlight on drone warfare in Ukraine** has validated the concept that swarms of cheap drones can have outsized impact – which supports funding for companies like AVAV and KTOS. The U.S. DoD has indeed accelerated these programs (e.g. the USAF's CCA program where awards may start by 2024-25). Both AVAV and KTOS stand to benefit if they secure roles in this new paradigm. Even aside from that, AVAV's core drone business is growing with or without cutting-edge AI, and Kratos has other niche programs (e.g. satellite communications tech, which could incorporate AI for network optimization).

Because these are smaller, more speculative, we keep them at a combined ~2%. Their stock movements can be volatile (AVAV spiked in 2022–23 due to Ukraine demand, then cooled; KTOS has traded range-bound waiting for a catalyst).

Recommendation: Continue to hold AVAV & KTOS at a modest allocation (1% each). This provides the portfolio a toe-hold in **AI-driven defense** – a potential right-tail outcome where, if an arms race intensifies, these stocks could multiply. The risk of loss is limited by the small size, and they somewhat hedge the scenario of heightened geopolitical conflict (which could hurt other parts of the portfolio but likely boosts defense names). Our internal review even contemplated a larger "AI-Enabled Defense" sleeve of up to 5–7.5%[122][124]. We have not implemented that fully, perhaps due to waiting for clearer signals. One could argue as confidence grows in this theme (say, Kratos wins a big CCA contract), we might **increase these or add another defense name**. For example, one might consider a larger defense prime like **Lockheed Martin or Northrop** which are also investing in autonomy – but those are much more diversified and not pure AI plays. Sticking with AVAV and KTOS targets the pure theme. So we will keep them and adjust if events dictate (e.g. if both perform very well and grow to, say, 4% combined, we might let them run or trim slightly depending on risk appetite).

In short, **this sleeve is a small bet with high optionality**. It fits the portfolio's broader strategy to hedge against geopolitical tail risks (if conflicts escalate, these could offset declines elsewhere). The current sizing is fine; no changes needed immediately, but an eye kept on developments in military AI adoption.

Bio-Risk Hedge – **Moderna (1%)**

The Bio-Risk "insurance" position is **Moderna (MRNA) 1%**, included as a hedge against the scenario where **AI advancements inadvertently increase biological risks (like engineered pandemics)**. The rationale is that if a new pathogen emerges (either naturally or via misuse of AI in bioengineering), a vaccine maker like Moderna could see outsized gains (as happened in 2020). Moderna's mRNA platform is considered a rapid response tool for novel diseases – essentially **an "anti-fragile" asset in a world of biotech risk**.

While this is not a core AI infrastructure holding, it's part of building a robust portfolio that can withstand second-order effects of AI. The **AI-2027 scenario analysis** explicitly raised the concern of AI enabling bad actors to create pathogens, i.e. "AI-enabled bioterrorism"[125]. If such a tail event occurred, much of our portfolio might suffer (market downturn, etc.), but **Moderna would likely soar** as governments pour resources into vaccines. Thus, the 1% position acts as a form of **tail-risk hedge** (small cost if nothing happens, big payoff if it does).

Current status: Moderna's stock has cooled off since the COVID vaccine boom – it went from over $400 at peak down to around the $100 range by 2025 as COVID vaccine sales declined. The company is now advancing a pipeline of other vaccines (RSV, flu combo shots, cancer vaccines in trials). It has huge cash reserves from its COVID profits to fund R&D. In an endemic world, its revenue is stabilizing, but any new pandemic or health crisis would dramatically change its fortunes.

Outlook: We don't expect Moderna to be a huge driver of returns absent a new pandemic – in fact, it might drift or trade range-bound as the market waits for new product successes. And that's fine for our purposes; we're not holding it for ordinary growth, but for **the extreme scenario that would otherwise blindside an AI-heavy portfolio**. If one wanted to refine this hedge, they could include other biotech or diagnostic companies, but Moderna is a straightforward, liquid proxy with proven capability (the fastest COVID vaccine).

Recommendation: Keep Moderna at 1%. It's a cheap insurance policy. There's no strong reason to change this allocation now – the "bet" hasn't paid off (which is good, meaning no devastating pandemic occurred). If over time Moderna's prospects in non-COVID areas improve significantly (say their cancer vaccine shows great results), then Moderna might warrant holding even on its own merits. That would be a bonus. If nothing notable happens, the 1% drag (if any) is minimal. We definitely wouldn't eliminate it because that would remove a thoughtful hedge. As internal strategy notes put it: *the threat of AI-enabled biohazards is credible, and the portfolio impact would be asymmetric – including a small allocation to Moderna is prudent as an insurance line*[125][126].

No alternative stock is as fitting (perhaps BioNTech or Pfizer could be used, but Moderna has the purest mRNA leverage). We will hold MRNA and hope we never need it to spike – but if we do, it will cushion the blow.

(Side note: We structured this not in the main "AI buildout" section but as a separate hedge, which is why it remains just 1%. We will revisit sizing if global bio risk seems to rise, but for now, 1% is sufficient.)

Strategic Materials – **MP Materials (3%)**

MP Materials (MP) is included to hedge and benefit from the **materials supply chain constraints** that could arise as AI (and related industries like EVs, renewables, defense) boom. MP owns the Mountain Pass mine in California, the only significant rare earth mine in the U.S., producing Neodymium-Praseodymium (NdPr) which is crucial for permanent magnets used in motors, generators, and many high-tech devices.

Why rare earths in an AI fund? Because advanced compute and infrastructure aren't just about silicon; they also rely on *physical materials* (metals, minerals). Rare earth magnets, for example, are used in the high-power electric motors of data center cooling systems (fans, pumps) and in wind turbines that generate clean power. Rare earths are also critical in satellites, sensors, and military hardware (all of which intersect with AI in various ways). Currently, **China dominates rare earth refining (>80% of supply)** and has shown willingness to weaponize that dominance (they've threatened export restrictions). If U.S.-China tensions escalate (not unlikely in a fast AI race scenario), rare earths could become a choke point – China could limit exports, causing prices to spike and Western supply to scramble[127][128]. MP Materials is effectively a *strategic asset* for the U.S. to secure its own supply of these minerals. The portfolio added MP as a 3% position recognizing that **critical minerals are an underappreciated Achilles heel** for tech supply chains[129][128]. By owning MP, we hedge against that risk (MP would benefit from higher rare earth prices or government support for domestic production).

Current status of MP: The Mountain Pass mine produces a concentrate that is still mostly shipped to China for processing (MP is building its own processing capacity, expected to start refining NdPr in the U.S. around 2025). Once that comes online, MP's value-add and margins should improve. They also have a contract to supply General Motors with magnets (meaning they are integrating down the chain to magnets manufacturing in a JV). Financially, MP has been profitable in recent years but saw some margin pressure as rare earth prices fluctuated (NdPr prices peaked in 2022 then softened). The stock is somewhat speculative, moving with commodity cycles and geopolitics. It's well-capitalized though, and the U.S. government has given it grants to help build refining facilities.

Outlook: We think **MP remains a solid hold as a strategic hedge**. If tensions with China rise, MP could see a surge (both in stock and possibly direct policy support). Even without conflict, the world's electrification (EVs, etc.) is increasing rare earth demand structurally. AI specifically might not drive rare earth demand as directly as it does chips or power, but indirectly (via more EVs, drones, robots – all needing magnets) it has an effect. Also, defense applications (like precision-guided munitions, which use rare earth components) could become more important – and AI escalation could coincide with defense escalation.

Recommendation: Keep MP at 3%. This size is enough to matter if the thesis plays out (e.g. if rare earths become "the new oil" in a trade war, MP could potentially multi-bag, making a real contribution at 3%). At the same time, if nothing dramatic happens, 3% in a mining stock won't significantly drag on overall performance (MP can be volatile, but it's not excessively large in the portfolio). No clear alternative public company fits this niche better – most rare earth or critical mineral plays are non-US or not pure-play. We note that MP is one of few ways to invest in the **"materials bottleneck"** for tech, and internal analysis indeed flagged critical minerals dependency as a severe risk to the AI thesis (e.g. reliance on China for refined materials)[130][128]. Owning MP is our way to address that risk.

We will monitor MP's operational progress (if they successfully start refining and magnet production, it will de-risk their business). If, say, MP were to falter or if another domestic rare earth company emerged and looked better, we could reallocate. But currently MP is essentially *the* U.S. rare earth champion, so we stick with it.

Now that we've reviewed all positions, the question remains: **Is the current allocation still optimal for maximizing future returns given the thesis and timeline?** And if not, what tweaks or new opportunities should be considered?

Portfolio Outlook and Recommendations

Having analyzed each component against the present state of AI infrastructure and industry trends, **the overall portfolio appears robust and well-aligned with the thesis**. The core idea – **invest in the bottlenecks** – is more valid than ever as evidence mounts of supply-demand imbalances in chips, memory, power, etc. Each major holding (TSMC, ASML, CEG, NVDA, etc.) has, if anything, **increased in strategic importance** over the past year. The portfolio also smartly covers a broad range of sectors (from semiconductors to utilities), providing diversification across different types of risk.

We find the current allocation to be largely optimal, with a few **minor adjustments** and **watchpoints** to discuss:

- **Stay Overweight Foundry, WFE, Power, and Cooling:** These are the high-conviction bottlenecks (and indeed the largest weights in the fund: TSMC 12%, ASML/AMAT 12%, CEG/PWR 18%, VRT/DELL 13%). All signals indicate these are still the pinch points in the AI race. We recommend **no reduction** in these; if anything, consider using future contributions or trims elsewhere to **top up these positions on dips**. For example, if market volatility hits TSMC or ASML stock, adding a percent or two more could be very rewarding, as their fundamental positions are entrenched. The same goes for Constellation or Vertiv – their recent performance affirms their thesis, and one should feel comfortable letting them constitute a big share of the portfolio. These holdings are the "spine" of the strategy, and we reaffirm them as such.

- **Slight Rebalancing Among Smaller Positions:** Among the 1–3% allocation names, we suggest a **few tweaks**:

- **Increase CrowdStrike (CRWD) from 2% to ~3-4%** on the next rebalancing. Rationale: The AI-driven threat environment is intensifying, and CrowdStrike's niche is vital and growing[102][105]. It's executing well (20%+ growth) and is more directly aligned with AI security needs than any competitor. This would strengthen the portfolio's coverage of digital risks, which complements the physical bottleneck focus.
- **Consider raising AI-Defense (AVAV+KTOS combined) from 2% closer to ~4-5%** if geopolitical tensions and defense AI programs continue accelerating. The internal recommendation was a 5–7.5% sleeve[122] – we haven't implemented fully, but we should be prepared to increase if, for instance, the U.S. awards major contracts to these firms (which would both boost their prospects and signal the dawn of the era of AI warfare). For now, one could inch to maybe 3% each or add a third name. Another approach is to keep them small but set aside a bit of cash to opportunistically add on news. In any case, we lean towards **not cutting** them; if anything, use positive developments as a cue to double down. This area could become a major theme by 2027 as nations view AI as a defense differentiator.
- **Maintain or trim Palantir at 2% depending on entry price:** Palantir's inclusion is strategic, but given its run-up earlier, it might oscillate. We bought in low, so currently we hold. If Palantir stock were to irrationally spike far above fundamentals on AI hype (say it doubled quickly without new earnings to back it), we might trim a little and reallocate to higher conviction core names. Conversely, on dips we'd top it back up. The target remains ~2%. So this is more about tactical management than strategic allocation change.
- **Keep MP Materials at 3%** as our strategic resource hedge. If new U.S. policies (like Defense Production Act for minerals) benefit MP, we reap rewards. If not, 3% exposure is manageable. We would re-evaluate only if MP faces some company-specific setback (none apparent now). We might also keep an eye on other "chokehold materials" (e.g. if something like high-grade graphite or silicon carbide becomes a serious pinch, though opportunities to invest directly in those are limited). MP currently suffices.

- **Watch for New Opportunities:** The current holdings cover the bases well, but we remain open to **adding a new position if a glaring opportunity arises that fits the thesis**. A few areas on our radar:

- **EDA – Cadence (CDNS):** We have Synopsys; adding Cadence (the other duopoly member) would double exposure to this very secure bottleneck[71]. At times one may outperform the other. Right now, Synopsys alone is fine, but if market pullback made valuations compelling, owning both could slightly reduce single-company risk in EDA and capture the full duopoly economics. Not a high priority, but noted.

- **Cloud/Challenger AI compute providers:** We intentionally avoided the big cloud stocks (GOOGL, MSFT, etc.) since they are not pure plays and are already held in the Continuity index fund portion. However, one intriguing name to watch is **Oracle (ORCL)**. Oracle has reinvented itself to provide cloud infrastructure targeted at AI workloads (e.g. partnering with NVIDIA for cloud GPUs) and has seen a surge in revenue because of it. It's sort of a "challenger hyperscaler" angle. That said, Oracle is still a broad enterprise software company; it doesn't neatly fit our "bottleneck" theme as it doesn't own unique physical assets, it's just competing well. Our research suggested adding ORCL to a watchlist but *not* the core portfolio, as it doesn't align with capacity-constrained producers[131][132]. We concur – no action now, but keep an eye in case Oracle's cloud pivot yields exceptional returns that outpace our picks.
- **Proprietary Data/Moat companies:** Another speculative theme is companies owning unique data sets that AI might need (for specialized training). This is a bit tangential to our focus on physical bottlenecks, and those companies (like RELX, Verisk, etc.) aren't clearly in high demand yet for AI training[133][134]. We agree with our earlier assessment: *monitor*, but no inclusion yet[133]. If, down the line, data licensing becomes a big business (say Verisk licenses insurance data for AI at scale), we could reconsider adding one. For now, they remain more "nice-to-have" than must-have in the portfolio.
- **Semiconductor sub-segments:** If we ever felt we wanted to broaden our semi exposure, we could add a Lam Research (for more memory equipment exposure) or an AMD (for more chip design exposure). At this point, we've rationalized not needing those due to overlaps (AMAT covers some of Lam's space; TSM covers benefit of AMD's growth). But we remain nimble – should Lam drop to very attractive levels or AMD prove it can grab, say, >20% AI GPU share consistently, we might take a small position opportunistically. Similarly, if **Intel** shows a credible turnaround in foundry or AI chips, we could consider it, though our plan there has been to use **options in the Moonshot sleeve** given Intel's high risk. That still seems prudent – Intel is making progress but is far from threatening TSMC or NVIDIA as of now. So no core position in INTC yet, consistent with "keep Intel as a moonshot, not core"[135].
- **Power Semiconductors (SiC/GaN):** One question is whether chips like silicon carbide (for power electronics) could become a chokepoint as data centers try to improve efficiency (SiC is used in high-efficiency power conversion). On watch, but not yet a clear bottleneck – multiple suppliers exist (Wolfspeed, ON Semi, etc.), and demand is high but more driven by EVs than AI specifically. We'll keep "power chips" on watchlist but not in portfolio unless a CoWoS-like scarcity appears (our earlier conclusion too)[136][137].
- **Network/Connectivity startups or alternatives:** We are quite set with Arista for networking. If Ethernet wins completely, Arista benefits; if not, our NVDA covers InfiniBand since NVIDIA owns that. One niche to watch is **co-packaged optics or optical interconnect startups**, but none are public, and any success they have

would likely funnel through either Arista or existing chip suppliers. So no action needed there for now.

- **Monitor Concentration and Correlation:** The portfolio is intentionally concentrated (top 5 positions make up ~50% of the fund). This has been by design to focus on highest convictions. It does mean volatility – as we saw, its beta to the semiconductor index (SOX) is high[138]. We are comfortable with that given the long horizon, but we periodically sanity-check that we're not missing a major area or overweighting a theme that might saturate early. Right now, each top holding addresses a distinct bottleneck, which is good. If any one of those theses shows signs of weakening (e.g., if a new technology alleviated one bottleneck), we would trim that in favor of the others. So far, none have – if anything all are reinforcing.

In conclusion, the Event Horizon Fund's AI sleeve is **well-positioned for the rapidly escalating AI arms race**. It is built on the insight that **when demand wildly outstrips supply, the suppliers hold the power – and profit**. Our review finds that dynamic very much in play across foundry capacity[10], advanced chips & memory[14][53], and even power grids[16]. The portfolio's focus on those areas rather than just chasing AI software hype sets it up for durable returns, in our view.

We endorse the current allocations with the minor adjustments noted (slightly higher weights in security and possibly defense, funded by new cash or trimming exuberant names if needed). We also emphasize **continued vigilance**: this is a fast-moving industry, so we will keep updating our situational awareness. The core strategy, however, appears sound and validated by data[13][59]. Thus, we conclude that the portfolio is **broadly optimal for maximizing future returns** given the thesis, and if maintained and tuned as suggested, it should capture outsized gains from the AI build-out through 2027 and beyond, while remaining resilient against associated risks.

Sources:

- TSMC's widening lead in 3 nm yields and record ~70% foundry market share[29][31].
- Chronic chip supply/demand gap (AI chips 80% CAGR vs wafers ~16% CAGR) and CoWoS packaging constraints sustaining TSMC's pricing power[4][10].
- SK Hynix, Samsung, Micron HBM market shares (57%, 27%, 16%) and Micron's HBM3E ramp capturing share[14][53].
- NVIDIA's ~80–90% AI accelerator share and CUDA software moat; AMD's emergence but expanding the overall market[36][39].
- Broadcom's $10B partnership with OpenAI to develop a custom AI chip (co-developed, fabbed at TSMC by 2026)[45].
- Vertiv's near-#1 position in data center power/cooling and **45× capacity expansion in liquid cooling units** to meet AI cluster demand[24][25].

- Dell's record $12.9B quarter (+69% YoY) in Servers & Networking and $20B+ annual run-rate for AI infrastructure, with ~$12B backlog for AI servers[88][89].
- Meta's **20-year nuclear energy deal** with Constellation, illustrating the repricing of existing nuclear as prime AI data center power source[21][22].
- The forecasted **50 GW U.S. AI datacenter power demand by 2027** and data centers' share of power use doubling by end of decade[16][139].
- Shift in AI networking: **Ethernet rapidly displacing InfiniBand**; Arista as the leader in high-performance cloud switching, optimized for AI clusters[15][76].
- CrowdStrike's focus on AI-driven threats and use of AI for defense (AI Security Posture Management, etc.), versus traditional firewalls becoming less relevant for insider AI threats[101][102].
- Palantir's unique role as an "AI operating system" with deep government integration and increasing enterprise adoption (mentions on earnings calls quadrupling)[114][116].
- Recommendation excerpts from internal analysis corroborating portfolio adjustments: e.g. boosting conviction in CEG (power bottleneck)[98], preferring CRWD over PANW for AI-era security[101], maintaining NVIDIA despite custom ASICs[47], and hedging bio-risk with Moderna[125].

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